

# Drummond B. Fielding

## Curriculum Vitae

Assistant Professor

Department of Physics, New York University, Office 1053,  
726 Broadway, New York, NY 10003

✉ [drummondfielding@gmail.com](mailto:drummondfielding@gmail.com) [dfelding14.github.io](https://github.com/dfelding14)  [arXiv](#)  [ORCID](#)

## Education

PhD 2018, MA 2014, Astrophysics, University of California, Berkeley. *Advisor: Eliot Quataert*  
BS, BA 2012, Physics, Mathematics, Johns Hopkins University.

## Professional Appointments

|                                                                                                            |           |
|------------------------------------------------------------------------------------------------------------|-----------|
| Assistant Professor, New York University, Department of Physics, Center for Cosmology and Particle Physics | 2025–     |
| Associate Research Scientist, Flatiron Institute, Center for Computational Astrophysics                    | 2025–     |
| Assistant Professor, Cornell University, Department of Astronomy                                           | 2024–2025 |
| Flatiron Research Fellow, CCA                                                                              | 2018–2024 |
| Visiting Scholar, Indian Institute of Science                                                              | 2016      |
| NSF & Berkeley Graduate Research Fellow, U.C., Berkeley                                                    | 2012–2018 |

## Research interests

|                      |                       |                       |
|----------------------|-----------------------|-----------------------|
| fluid dynamics       | galactic winds        | stellar+BH feedback   |
| plasma physics       | circumgalactic medium | magnetic reconnection |
| galaxy formation     | magnetic turbulence   | geom. measure theory  |
| cosmic ray transport | multiphase flows      | machine learning      |

## Grants

*The Geometry of Flows*, Simons Foundation, PI, 2025–2030  
*Pushing the Frontier of Cosmic Ray Transport in Interstellar Turbulence*, DOE INCITE, PI, 2025–2027  
*Resolving Cosmic Ray Transport by pushing the Frontier of MHD Turbulence*, DOE INCITE, PI, 2024  
*Illuminating the Interplay of Multiphase Galactic Winds and the CGM*, NSF AAG, PI  
*Mocking Galactic Winds: A new approach to constraining feedback by modeling the UV spectra of galaxies*, Legacy HST-AR, Co-PI, 2025–2027  
*Galactic Winds Unveiled: Leveraging Cloud Simulations with Radiative Transfer to Constrain Feedback*, Legacy HST-AR, Co-PI, 2024–2026  
*Resolving Turbulent Multiphase Gas Dynamics*, EURO HPC, Co-PI, 2023–2025  
*AthenaK: a Performance Portable Simulation Infrastructure for Computational Astrophysics*, NASA Support for Open-Source Tools, Frameworks, and Libraries, Co-I, 2025–2027  
*DUVET: Mapping Outflows and the Inner CGM in Starbursting Disks*, KCWI, Co-I, 2020–  
Visiting scholar grant to work Indian Institute of Science with Prof. Prateek Sharma, APS-IUSSTF, PI, 2016  
*Towards an Understanding of the Origin of OVI in the Circumgalactic Medium*, HST AR, Co-I, 2017  
*The Physics of Supernova Feedback: Global 3D Simulations of Galactic Disks*, NSF XSEDE, Co-I, 2016–2018  
*Conduction, Convection, and Thermal Instability in Hot Halos*, NSF XSEDE, Co-I, 2016–2017

## Advising

---

[Brent Tan](#) — Postdoc — Cornell University and NYU  
[Hitesh Das](#) — Postdoc — NYU  
[Alex Pendris](#) — Graduate Student — NYU  
[Nana Ama Darpaah](#) — Masters Student — NYU  
[Lillion Shi](#) — Undergraduate — NYU  
[Yongqi Zhang](#) — Graduate Student — Cornell University  
[Bodong Liu](#) — Undergraduate Student — Cornell University  
[Iryna Butsky](#) — Graduate Student — University of Washington + CCA predoctoral fellow → Hubble Fellow at Stanford  
[Viraj Pandya](#) — Graduate Student — UC Santa Cruz + CCA predoctoral fellow → Hubble Fellow at Columbia  
[Matthew Abruzzo](#) — Graduate Student — Columbia → Postdoc at UPitt  
[Anthony Chow](#) — Graduate Student — Columbia  
[Zirui Chen](#) — Undergraduate Student — Columbia → Grad Student at UCSB  
[Minami Roy](#) — Graduate Student — Raman Research Institute + CCA predoctoral fellow → Postdoc at OSU  
[Mitali Damle](#) — Graduate Student — University of Potsdam → Postdoc at NYU Abu Dhabi

## Professional services & teaching

---

Astro 1101: From New Worlds to Black Holes — Cornell University — 200+ undergraduates Fall 2024  
 Astro 6590: Galaxies and the Universe — Cornell University — graduate level Spring 2025  
 Founder and organizer of the *New York Area Fluid Dynamics Meeting*, a cross-disciplinary monthly meeting of local fluid dynamacists, 2020–  
 Research Mentor, *Simons-National Society of Black Physicists Scholars Program*, 2023  
[Flatiron Pre-Doctoral Program](#) graduate student mentor, 2019–2024  
 Referee: MNRAS, ApJ, ApJL, PRL  
 Proposal evaluation panelist: NSF (2020), NASA HST (2024)  
 Summer research mentor for the *Cornell Nexus Scholars Program*, 2025–

## Honors and awards

---

|                                                            |           |
|------------------------------------------------------------|-----------|
| Outstanding Graduate Student Instructor Award              | 2014      |
| NSF Graduate Research Fellowship                           | 2014–2017 |
| Berkeley Graduate Fellowship                               | 2012–2014 |
| Donald E. Kerr Award for Outstanding Physics Undergraduate | 2012      |
| The Phi Beta Kappa Society                                 | 2012      |

## Public outreach

---

Founder and leader of *The Astro Club*. Monthly meetings 2020–2023. [Democracy Prep Charter Middle School](#), Harlem, NY.  
 Astronomy career day, [Democracy Prep Charter Middle School](#), Harlem, NY, 2019  
 Taught a month-long series of astronomy classes to 2nd and 3rd grade students, 2017. North Oakland Community Charter School, Oakland, CA  
 UC Berkeley Astronomy Department Public Liaison, 2015  
 East Bay Astronomical Society Public Lecture, 2014, Chabot Space and Science Center  
[The Berkeley Compass Project](#) Summer Evening Instructor, 2013

## Selected recent presentations

---

Endowed Colloquium, Niels Bohr Lecture, Niels Bohr Institute, Copenhagen (10/2024)  
Seminar, Institute for Advanced Studies, Princeton (10/2024)  
Colloquium, Flatiron Institute, Center for Computational Astrophysics (2/2024)  
Invited Review, *New simulations for new problems in galaxy formation*, conference, Paris (12/2023)  
Invited Talk, Salpeter Workshop on the Interstellar Medium, Cornell (12/2023)  
Seminar, Computational Research in Boston and Beyond (CRIBB), MIT (12/2023)  
Colloquium, Johns Hopkins University, Physics and Astronomy Department (11/2023)  
Colloquium, Columbia University, Astronomy Department (10/2023)  
Invited talk, *Interfaces and Mixing in Fluids, Plasmas, and Materials*, conference, KITP (9/2023)  
Colloquium, Michigan State University, Physics and Astronomy Department (10/2023)  
Invited Review, *MIST2023: Cosmic turbulence and Magnetic fields: physics of baryonic matter across time and scales*, conference (9/2023)  
Invited talk, CGM@ND, conference (9/2023)  
Invited talk, *New Views on Feedback & the Baryon Cycle in Galaxies*, conference (8/2023)  
Invited talk, *Modeling multiphase astrophysical systems*, conference (5/2023)  
Invited talk, *Jim's Fest/Athena++*, conference (5/2023)  
Colloquium, Yale University, Astronomy Department (2/2023)  
Colloquium, Cornell University, Astronomy Department (1/2023)  
Invited Astrophysics Seminar, UIUC (11/2022)  
Joint Astrophysics Colloquium, Pitt-CMU (10/2022)  
Invited talk, *What Matter(s) Around Galaxies*, conference (9/2022)  
Invited talk, *Ringberg Computational Galaxy Formation 2022*, conference (4/2022)  
Invited talk, *Wolfe Symposium*, conference (3/2022)  
Invited Seminar, Theoretical AstroPhysics Including Relativity & Cosmology, Caltech (12/2021)  
Astronomy Department seminar, Racah Institute of Physics, Hebrew University of Jerusalem (12/2021)  
Invited Talk, *Baltimore Wind Workshop*, Johns Hopkins University (8/2021)  
Astrophysics Colloquium, UCSD-SDSU (4/2021)  
[Keynote speaker](#), KITP workshop *Fundamentals of Gaseous Halos* (1/2021)  
Invited talk, KITP conference *On the Origin, Nature, and Mixing of Multiphase Gas in Astrophysics* (10/2020)  
Gaggle Seminar, Carnegie Institution for Science (8/2020)  
Invited talk, CLEARNESS conference IAP (6/2020)  
CGI Colloquium, U.C. Santa Cruz (4/2020)  
Contributed Talk, Universality of Turbulence Conference, Flatiron Institute, NY (12/2019)  
SFIR Seminar, Princeton University (11/2019)  
Astro Seminar, NYU CCPP, New York, NY (10/2019)  
Invited Talk, CGM conference, Berlin, Germany (10/2019)  
Invited seminar, Turbulence workshop, Aspen Center for Physics (6/2019)  
Invited Talk, *athena++* conference, Las Vegas, NV (3/2019)  
Invited Talk, CGM conference, Northwestern, Chicago, IL (8/2018)  
Invited Talk, CGM/DLA conference, Big Sur, CA (3/2018)  
TAPIR Seminar, Caltech, Pasadena, CA (11/2017)  
Astronomy Colloquium, Raman Research Institute (12/2016)  
Astronomy Department Colloquium, Indian Institute of Science (10/2016)

## Publications— *ADS search* — *Google Scholar* — *ORCID*

total publication count: 84 — citations: 3728 — h-index: 36 (12 Jul 2026)

\* indicates  $\geq 100$  citations — ‡ graduate student-led — § undergraduate-led — † postdoc-led

### First Author

- 8 **Fielding, Drummond B.**; Ripperda, Bart; Philippov, Alexander A., [Plasmoid Instability in the Multiphase Interstellar Medium](#), *ApJ*, 949, L5, 2023 ([arXiv:2211.06434](#)) [[43 citations](#)]
- \*7 **Fielding, Drummond B.**; Bryan, Greg L., [The Structure of Multiphase Galactic Winds](#), *ApJ*, 924, 82, 2022 ([arXiv:2108.05355](#)) [[157 citations](#)]
- 6 **Fielding, Drummond B.**; Tonnesen, Stephanie; DeFelippis, Daniel; Li, Miao *et al.*, [First Results from SMAUG: Uncovering the Origin of the Multiphase Circumgalactic Medium with a Comparative Analysis of Idealized and Cosmological Simulations](#), *ApJ*, 903, 32, 2020 ([arXiv:2006.16316](#)) [[76 citations](#)]
- \*5 **Fielding, Drummond B.**; Ostriker, Eve C.; Bryan, Greg L.; Jermyn, Adam S., [Multiphase Gas and the Fractal Nature of Radiative Turbulent Mixing Layers](#), *ApJ*, 894, L24, 2020 ([arXiv:2003.08390](#)) [[187 citations](#)]
- \*4 **Fielding, Drummond B.**; Quataert, Eliot; Martizzi, Davide, [Clustered supernovae drive powerful galactic winds after superbubble breakout](#), *MNRAS*, 481, 3325, 2018 ([arXiv:1807.08758](#)) [[165 citations](#)]
- 3 **Fielding, Drummond B.**; Quataert, Eliot; Martizzi, Davide; Faucher-Giguère, Claude-André, [How supernovae launch galactic winds?](#), *MNRAS*, 470, L39, 2017 ([arXiv:1704.01579](#)) [[97 citations](#)]
- \*2 **Fielding, Drummond B.**; Quataert, Eliot; McCourt, Michael; Thompson, Todd A., [The impact of star formation feedback on the circumgalactic medium](#), *MNRAS*, 466, 3810, 2017 ([arXiv:1606.06734](#)) [[183 citations](#)]
- \*1 **Fielding, Drummond B.**; McKee, Christopher F.; Socrates, Aristotle; Cunningham, Andrew J. *et al.*, [The turbulent origin of spin-orbit misalignment in planetary systems](#), *MNRAS*, 450, 3306, 2015 ([arXiv:1409.5148](#)) [[102 citations](#)]

### Second Author

- 21 Lancaster, Lachlan; **Fielding, Drummond B.**; Mohapatra, Rajsekhar; Bryan, Greg L., [The Origin of Da Scaling: Suppressed Cooling in Fast-Cooling Mixing Layers](#), *arXiv e-prints*, 2026 ([arXiv:2606.04093](#))
- 20 Xu, Xinfeng; **Fielding, Drummond B.**; Heckman, Timothy; Bryan, Greg L. *et al.*, [Resolving the Unresolved Galactic Winds in Multiphase Models. I. Methodology and Application](#), *ApJ*, 1003, 155, 2026 ([arXiv:2605.01105](#))
- 19 Kempfski, Philipp; **Fielding, Drummond B.**; Quataert, Eliot; Ewart, Robert J. *et al.*, [Self-similar Cosmic-Ray Transport in High-resolution Magnetohydrodynamic Turbulence](#), *ApJ*, 994, L49, 2025 ([arXiv:2507.10651](#)) [[13 citations](#)]
- 18 Smith, Matthew C.; **Fielding, Drummond B.**; Bryan, Greg L.; Bennett, Jake S. *et al.*, [ARKENSTONE - II. A model for unresolved cool clouds entrained in galactic winds in cosmological simulations](#), *MNRAS*, 535, 3550, 2024 ([arXiv:2408.15321](#)) [[23 citations](#)]
- 17 Stern, Jonathan; **Fielding, Drummond B.**; Hafen, Zachary; Su, Kung-Yi *et al.*, [Correction to: accretion onto disc galaxies via hot and rotating CGM inflows](#), *MNRAS*, 532, 995, 2024
- 16 Stern, Jonathan; **Fielding, Drummond B.**; Hafen, Zachary; Su, Kung-Yi *et al.*, [Accretion onto disc galaxies via hot and rotating CGM inflows](#), *MNRAS*, 530, 1711, 2024 ([arXiv:2306.00092](#)) [[50 citations](#)]
- ‡15 Abruzzo, Matthew W.; **Fielding, Drummond B.**; Bryan, Greg L., [Taming the TuRMoiL: The Temperature Dependence of Turbulence in Cloud-Wind Interactions](#), *ApJ*, 966, 181, 2024 ([arXiv:2210.15679](#)) [[36 citations](#)]
- †14 Tan, Brent; **Fielding, Drummond B.**, [Cloud atlas: navigating the multiphase landscape of tempestuous galactic winds](#), *MNRAS*, 527, 9683, 2024 ([arXiv:2305.14424](#)) [[57 citations](#)]
- 13 Smith, Matthew C.; **Fielding, Drummond B.**; Bryan, Greg L.; Kim, Chang-Goo *et al.*, [ARKENSTONE - I. A novel method for robustly capturing high specific energy outflows in cosmological simulations](#), *MNRAS*, 527, 1216, 2024 ([arXiv:2301.07116](#)) [[41 citations](#)]
- 12 Kempfski, Philipp; **Fielding, Drummond B.**; Quataert, Eliot; Galishnikova, Alisa K. *et al.*, [Cosmic ray transport in large-amplitude turbulence with small-scale field reversals](#), *MNRAS*, 525, 4985, 2023 ([arXiv:2304.12335](#)) [[77 citations](#)]
- ‡11 Pandya, Viraj; **Fielding, Drummond B.**; Bryan, Greg L.; Carr, Christopher *et al.*, [A Unified Model for the Coevolution of Galaxies and Their Circumgalactic Medium: The Relative Roles of Turbulence and Atomic Cooling Physics](#), *ApJ*, 956, 118, 2023 ([arXiv:2211.09755](#)) [[48 citations](#)]

- ‡10 Abruzzo, Matthew W.; **Fielding, Drummond B.**; Bryan, Greg L., [TuRMoiL of Survival: A Unified Survival Criterion for Cloud-Wind Interactions](#), arXiv e-prints, 2023 ([arXiv:2307.03228](#)) [[21 citations](#)]
- §9 Chen, Zirui; **Fielding, Drummond B.**; Bryan, Greg L., [The Anatomy of a Turbulent Radiative Mixing Layer: Insights from an Analytic Model with Turbulent Conduction and Viscosity](#), *ApJ*, 950, 91, 2023 ([arXiv:2211.01395](#)) [[20 citations](#)]
- ‡8 Orr, Matthew E.; **Fielding, Drummond B.**; Hayward, Christopher C.; Burkhart, Blakesley, [Bursting Bubbles: Feedback from Clustered Supernovae and the Trade-off Between Turbulence and Outflows](#), *ApJ*, 932, 88, 2022 ([arXiv:2109.14656](#)) [[43 citations](#)]
- ‡7 Orr, Matthew E.; **Fielding, Drummond B.**; Hayward, Christopher C.; Burkhart, Blakesley, [Bursting Bubbles: Clustered Supernova Feedback in Local and High-redshift Galaxies](#), *ApJ*, 924, L28, 2022 ([arXiv:2109.14626](#)) [[16 citations](#)]
- 6 Stachenfeld, Kimberly; **Fielding, Drummond B.**; Kochkov, Dmitrii; Cranmer, Miles *et al.*, [Learned Coarse Models for Efficient Turbulence Simulation](#), arXiv e-prints, 2021 ([arXiv:2112.15275](#)) [[76 citations](#)]
- \*‡5 Pandya, Viraj; **Fielding, Drummond B.**; Anglés-Alcázar, Daniel; Somerville, Rachel S. *et al.*, [Characterizing mass, momentum, energy, and metal outflow rates of multiphase galactic winds in the FIRE-2 cosmological simulations](#), *MNRAS*, 508, 2979, 2021 ([arXiv:2103.06891](#)) [[164 citations](#)]
- \*‡4 Butsky, Iryna S.; **Fielding, Drummond B.**; Hayward, Christopher C.; Hummels, Cameron B. *et al.*, [The Impact of Cosmic Rays on Thermal Instability in the Circumgalactic Medium](#), *ApJ*, 903, 77, 2020 ([arXiv:2008.04915](#)) [[105 citations](#)]
- 3 Stern, Jonathan; **Fielding, Drummond B.**; Faucher-Giguère, Claude-André; Quataert, Eliot, [The maximum accretion rate of hot gas in dark matter haloes](#), *MNRAS*, 492, 6042, 2020 ([arXiv:1909.07402](#)) [[78 citations](#)]
- \*2 Stern, Jonathan; **Fielding, Drummond B.**; Faucher-Giguère, Claude-André; Quataert, Eliot, [Cooling flow solutions for the circumgalactic medium](#), *MNRAS*, 488, 2549, 2019 ([arXiv:1906.07737](#)) [[104 citations](#)]
- \*1 Martizzi, Davide; **Fielding, Drummond B.**; Faucher-Giguère, Claude-André; Quataert, Eliot, [Supernova feedback in a local vertically stratified medium: interstellar turbulence and galactic winds](#), *MNRAS*, 459, 2311, 2016 ([arXiv:1601.03399](#)) [[110 citations](#)]

## Co-author

- 55 Sankar, Sriram; Stern, Jonathan; Power, Chris; Catinella, Barbara *et al.* (incl. **DBF**), [Hot accretion onto spiral galaxies: the origin of extended and warped H I discs](#), *MNRAS*, 549, stag1059, 2026 ([arXiv:2511.03793](#)) [[4 citations](#)]
- 54 Butsky, Iryna S.; Redshaw, Caleb; Lei, Minjie; Clark, Susan E. *et al.* (incl. **DBF**), [Phase-dependent magnetic coherence in the turbulent interstellar medium](#), arXiv e-prints, 2026 ([arXiv:2606.11323](#))
- 53 Chen, Zirui; Oh, S. Peng; **Fielding, Drummond B.**; Lancaster, Lachlan *et al.*, [Clumps in a Cocoon: Geometry and Mixing Set the Universal X-ray to H \$\alpha\$  Surface Brightness Ratio](#), arXiv e-prints, 2026 ([arXiv:2606.07741](#))
- 52 Lancaster, Lachlan; Mohapatra, Rajsekhar; **Fielding, Drummond B.**; Bryan, Greg L., [Ceci n'est pas une Couche de Mélange: The Meaning of Resolved Turbulent Radiative Mixing](#), arXiv e-prints, 2026 ([arXiv:2606.04087](#)) [[1 citations](#)]
- 51 Goldner, Roy; Stern, Jonathan; **Fielding, Drummond B.**; Faucher-Giguère, Claude-André *et al.*, [Accretion-driven turbulence in the circumgalactic medium](#), *MNRAS*, 548, stag717, 2026 ([arXiv:2510.27678](#)) [[5 citations](#)]
- 50 Sullivan, James M.; Bryan, Greg L.; Smith, Matthew C.; Bennett, Jake S. *et al.* (incl. **DBF**), [ArkenstoneBH. A model for high-specific energy black hole feedback in cosmological simulations](#), arXiv e-prints, 2026 ([arXiv:2605.03154](#)) [[1 citations](#)]
- 49 Chen, Zirui; Peng, Zixuan; Rubin, Kate H. R.; Heckman, Timothy M. *et al.* (incl. **DBF**), [Modelling emission-line surface brightness in a multiphase galactic wind: an O VI case study](#), *MNRAS*, 548, stag617, 2026 ([arXiv:2510.02443](#)) [[1 citations](#)]
- 48 Pandya, Viraj; Bryan, Greg L.; Makinen, T. Lucas; Gabrielpillai, Austen *et al.* (incl. **DBF**), [Introducing sapphire: Towards Hybrid Physics-Informed, Data-Driven Modeling of Galaxy Formation](#), arXiv e-prints, 2026 ([arXiv:2604.06318](#)) [[4 citations](#)]
- 47 Bennett, Jake S.; Smith, Matthew C.; **Fielding, Drummond B.**; Bryan, Greg L. *et al.*, [Correction to: Prevention is better than cure? Feedback from high specific energy winds in cosmological simulations with ARKENSTONE](#), *MNRAS*, 547, stag374, 2026
- 46 Stone, James M.; Mullen, Patrick D.; **Fielding, Drummond B.**; Grete, Philipp *et al.*, [AthenaK: A Performance-portable Version of the Athena++ Adaptive Mesh Refinement Framework](#), *ApJS*, 283, 27, 2026 ([arXiv:2409.16053](#)) [[80 citations](#)]
- 45 Ramesh, Rahul; Nelson, Dylan; **Fielding, Drummond B.**; Brüggén, Marcus, [Zooming in on the circumgalactic medium with GIBLE: Cloud-scale simulations with cosmological initial conditions](#), arXiv e-prints, 2026 ([arXiv:2602.23416](#)) [[3 citations](#)]
- 44 Burger, Jan D.; Springel, Volker; Ostriker, Eve C.; Kim, Chang-Goo *et al.* (incl. **DBF**), [Applying a star formation model calibrated on high-resolution interstellar medium simulations to cosmological simulations of galaxy formation](#), *MNRAS*, 544, 1390, 2025



- (arXiv:2502.13244) [10 citations]
- 43 Kakoly, Aharon; Stern, Jonathan; Faucher-Giguère, Claude-André; **Fielding, Drummond B. et al.**, Turbulence-dominated CGM: the origin of UV absorbers with equivalent widths of  $\sim 1 \text{ \AA}$ , MNRAS, 543, 3345, 2025 (arXiv:2504.17001) [18 citations]
  - 42 Lei, Minjie; Clark, S. E.; Morel, Rudy; Allys, E. et al. (incl. **DBF**), Neutral Gas Phase Distribution from H I Morphology: Phase Separation with Scattering Spectra and Variational Autoencoders, ApJ, 993, 4, 2025 (arXiv:2505.20407) [4 citations]
  - 41 Bennett, Jake S.; Smith, Matthew C.; **Fielding, Drummond B.**; Bryan, Greg L. et al., Prevention is better than cure? Feedback from high specific energy winds in cosmological simulations with ARKENSTONE, MNRAS, 543, 1456, 2025 (arXiv:2410.12909) [16 citations]
  - 40 Porter, Lori E.; Abruzzo, Matthew; Bryan, Greg L.; Putman, Mary et al. (incl. **DBF**), Simulating high-velocity clouds in the observational plane: an initial study with the Smith Cloud, MNRAS, 542, 1188, 2025 (arXiv:2506.00111)
  - 39 Steinwandel, Ulrich P.; Rennehan, Douglas; Orr, Matthew E.; **Fielding, Drummond B. et al.**, Pumping Iron: How Turbulent Metal Diffusion Impacts Multiphase Galactic Outflows, ApJ, 991, 16, 2025 (arXiv:2407.14599) [12 citations]
  - 38 Kempster, Philipp; Li, Dongzi; **Fielding, Drummond B.**; Quataert, Eliot et al., A Unified Model of Cosmic-Ray Propagation and Radio Extreme Scattering Events from Intermittent Interstellar Structures, ApJ, 990, L18, 2025 (arXiv:2412.03649) [11 citations]
  - 37 Mohapatra, Rajsekhar; Quataert, Eliot; **Fielding, Drummond B.**; Guo, Minghao, The Type Ia Supernova and Asymptotic Giant Branch Stellar Ejecta-regulated Interstellar Medium of Massive Galaxies, ApJ, 989, 103, 2025 (arXiv:2502.05329) [4 citations]
  - 36 Ramesh, Rahul; Nelson, Dylan; **Fielding, Drummond B.**; Brüggén, Marcus, Zooming in on the circumgalactic medium with GIBLE: Tracing the origin and evolution of cold clouds, A&A, 699, A10, 2025 (arXiv:2407.00172) [15 citations]
  - 35 Fisher, Deanne B.; Bolatto, Alberto D.; Chisholm, John; **Fielding, Drummond B. et al.**, JWST observations of starbursts: cold clouds and plumes launching in the M 82 outflow, MNRAS, 538, 3068, 2025 (arXiv:2405.03686) [26 citations]
  - 34 Peng, Zixuan; Martin, Crystal L.; Chen, Zirui; **Fielding, Drummond B. et al.**, Physical Origins of Outflowing Cold Clouds in Local Star-forming Dwarf Galaxies, ApJ, 981, 171, 2025 (arXiv:2412.05371) [9 citations]
  - 33 Reichardt Chu, Bronwyn; Fisher, Deanne B.; Chisholm, John; Berg, Danielle et al. (incl. **DBF**), DUVET: sub-kiloparsec resolved star formation driven outflows in a sample of local starbursting disc galaxies, MNRAS, 536, 1799, 2025 (arXiv:2402.17830) [27 citations]
  - 32 Qutob, Nadia; Emami, Razieh; Su, Kung-Yi; Smith, Randall et al. (incl. **DBF**), Observational Signatures of AGN Feedback in the Morphology and the Ionization States of Milky Way-like Galaxies, ApJ, 977, 72, 2024 (arXiv:2312.14809) [5 citations]
  - 31 Choi, Bo-Eun; Werk, Jessica K.; Tchernyshyov, Kirill; Prochaska, J. Xavier et al. (incl. **DBF**), Metallicity Mapping of the Ionized Diffuse Gas at the Milky Way Disk-Halo Interface, ApJ, 976, 222, 2024 (arXiv:2410.06286) [2 citations]
  - 30 Voit, G. Mark; Carr, Christopher; **Fielding, Drummond B.**; Pandya, Viraj et al., Equilibrium States of Galactic Atmospheres. II. Interpretation and Implications, ApJ, 976, 151, 2024 (arXiv:2406.07632) [13 citations]
  - 29 Voit, G. Mark; Pandya, Viraj; **Fielding, Drummond B.**; Bryan, Greg L. et al., Equilibrium States of Galactic Atmospheres. I. The Flip Side of Mass Loading, ApJ, 976, 150, 2024 (arXiv:2406.07631) [17 citations]
  - 28 Ohana, Ruben; McCabe, Michael; Meyer, Lucas; Morel, Rudy et al. (incl. **DBF**), The Well: a Large-Scale Collection of Diverse Physics Simulations for Machine Learning, arXiv e-prints, 2024 (arXiv:2412.00568) [16 citations]
  - 27 Perrotta, Serena; Coil, Alison L.; Rupke, David S. N.; Ning, Wenmeng et al. (incl. **DBF**), The Outflowing [O II] Nebulae of Compact Starburst Galaxies at  $z \sim 0.5$ , ApJ, 975, 263, 2024 (arXiv:2409.10013) [8 citations]
  - 26 Hassan, Sultan; Ostriker, Eve C.; Kim, Chang-Goo; Bryan, Greg L. et al. (incl. **DBF**), Toward Implementation of the Pressure-regulated, Feedback-modulated Model of Star Formation in Cosmological Simulations: Methods and Application to TNG, ApJ, 975, 151, 2024 (arXiv:2409.09121) [9 citations]
  - 25 Hummels, Cameron B.; Rubin, Kate H. R.; Schneider, Evan E.; **Fielding, Drummond B.**, CLOUDFLEX: A Flexible Parametric Model for the Small-scale Structure of the Circumgalactic Medium, ApJ, 972, 148, 2024 (arXiv:2311.05691) [23 citations]
  - 24 Kim, Chang-Goo; Ostriker, Eve C.; Kim, Jeong-Gyu; Gong, Munan et al. (incl. **DBF**), Metallicity Dependence of Pressure-regulated Feedback-modulated Star Formation in the TIGRESS-NCR Simulation Suite, ApJ, 972, 67, 2024 (arXiv:2405.19227) [20 citations]
  - 23 Su, Kung-Yi; Bryan, Greg L.; Hayward, Christopher C.; Somerville, Rachel S. et al. (incl. **DBF**), Unravelling jet quenching criteria across  $L^*$  galaxies and massive cluster ellipticals, MNRAS, 532, 2724, 2024 (arXiv:2310.17692) [22 citations]
  - 22 Hamel-Bravo, Magdalena J.; Fisher, Deanne B.; Berg, Danielle; Björngvinsson, Bjarki et al. (incl. **DBF**), DUVET: Resolved direct metallicity measurements in the outflow of starburst galaxy NGC 1569, MNRAS, 530, 3855, 2024 (arXiv:2404.04600) [19

citations]

- ‡21 Ramesh, Rahul; Nelson, Dylan; **Fielding, Drummond B.**; Brüggen, Marcus, [Zooming in on the circumgalactic medium with GIBLE. The topology and draping of magnetic fields around cold clouds](#), *A&A*, 684, L16, 2024 (arXiv:2404.01370) [18 citations]
- 20 Coil, Alison L.; Perrotta, Serena; Rupke, David S. N.; Lochhaas, Cassandra *et al.* (incl. **DBF**), [Ionized gas extends over 40 kpc in an odd radio circle host galaxy](#), *Nature*, 625, 459, 2024 (arXiv:2310.15162) [23 citations]
- 19 Roy, Manami; Su, Kung-Yi; Tonnesen, Stephanie; **Fielding, Drummond B.** *et al.*, [Seeding the CGM: how satellites populate the cold phase of milky way haloes](#), *MNRAS*, 527, 265, 2024 (arXiv:2310.04404) [20 citations]
- 18 Steinwandel, Ulrich P.; Kim, Chang-Goo; Bryan, Greg L.; Ostriker, Eve C. *et al.* (incl. **DBF**), [The Structure and Composition of Multiphase Galactic Winds in a Large Magellanic Cloud Mass Simulated Galaxy](#), *ApJ*, 960, 100, 2024 (arXiv:2212.03898) [56 citations]
- 17 McPherson, Daniel K.; Fisher, Deanne B.; Nielsen, Nikole M.; Kacprzak, Glenn G. *et al.* (incl. **DBF**), [DUVET survey: mapping outflows in the metal-poor starburst Mrk 1486](#), *MNRAS*, 525, 6170, 2023 (arXiv:2308.06918) [18 citations]
- ‡16 Carr, Christopher; Bryan, Greg L.; **Fielding, Drummond B.**; Pandya, Viraj *et al.*, [Regulation of Star Formation by a Hot Circumgalactic Medium](#), *ApJ*, 949, 21, 2023 (arXiv:2211.05115) [37 citations]
- 15 Reichardt Chu, Bronwyn; Fisher, Deanne B.; Bolatto, Alberto D.; Chisholm, John *et al.* (incl. **DBF**), [DUVET: Spatially Resolved Observations of Star Formation Regulation via Galactic Outflows in a Starbursting Disk Galaxy](#), *ApJ*, 941, 163, 2022 (arXiv:2211.02063) [18 citations]
- \*14 Hafen, Zachary; Stern, Jonathan; Bullock, James; Gurvich, Alexander B. *et al.* (incl. **DBF**), [Hot-mode accretion and the physics of thin-disc galaxy formation](#), *MNRAS*, 514, 5056, 2022 (arXiv:2201.07235) [110 citations]
- 13 Butsky, Iryna S.; Werk, Jessica K.; Tchernyshyov, Kirill; **Fielding, Drummond B.** *et al.*, [The Impact of Cosmic Rays on the Kinematics of the Circumgalactic Medium](#), *ApJ*, 935, 69, 2022 (arXiv:2106.14889) [34 citations]
- ‡12 Abruzzo, Matthew W.; Bryan, Greg L.; **Fielding, Drummond B.**, [A Simple Model for Mixing and Cooling in Cloud-Wind Interactions](#), *ApJ*, 925, 199, 2022 (arXiv:2101.10344) [64 citations]
- 11 Stern, Jonathan; Sternberg, Amiel; Faucher-Giguère, Claude-André; Hafen, Zachary *et al.* (incl. **DBF**), [Neutral CGM as damped Ly  \$\alpha\$  absorbers at high redshift](#), *MNRAS*, 507, 2869, 2021 (arXiv:2105.06489) [40 citations]
- \*10 Stern, Jonathan; Faucher-Giguère, Claude-André; **Fielding, Drummond B.**; Quataert, Eliot *et al.*, [Virialization of the Inner CGM in the FIRE Simulations and Implications for Galaxy Disks, Star Formation, and Feedback](#), *ApJ*, 911, 88, 2021 (arXiv:2006.13976) [144 citations]
- 9 Burkhart, B.; Appel, S. M.; Bialy, S.; Cho, J. *et al.* (incl. **DBF**), [The Catalogue for Astrophysical Turbulence Simulations \(CATS\)](#), *ApJ*, 905, 14, 2020 (arXiv:2010.11227) [30 citations]
- 8 Pandya, Viraj; Somerville, Rachel S.; Anglés-Alcázar, Daniel; Hayward, Christopher C. *et al.* (incl. **DBF**), [First Results from SMAUG: The Need for Preventative Stellar Feedback and Improved Baryon Cycling in Semianalytic Models of Galaxy Formation](#), *ApJ*, 905, 4, 2020 (arXiv:2006.16317) [54 citations]
- 7 Kim, Chang-Goo; Ostriker, Eve C.; **Fielding, Drummond B.**; Smith, Matthew C. *et al.*, [A Framework for Multiphase Galactic Wind Launching Using TIGRESS](#), *ApJ*, 903, L34, 2020 (arXiv:2010.09090) [69 citations]
- \*6 Kim, Chang-Goo; Ostriker, Eve C.; Somerville, Rachel S.; Bryan, Greg L. *et al.* (incl. **DBF**), [First Results from SMAUG: Characterization of Multiphase Galactic Outflows from a Suite of Local Star-forming Galactic Disk Simulations](#), *ApJ*, 900, 61, 2020 (arXiv:2006.16315) [147 citations]
- 5 Lochhaas, Cassandra; Bryan, Greg L.; Li, Yuan; Li, Miao *et al.* (incl. **DBF**), [Properties of the simulated circumgalactic medium](#), *MNRAS*, 493, 1461, 2020 (arXiv:1908.00021) [45 citations]
- 4 Martizzi, Davide; Quataert, Eliot; Faucher-Giguère, Claude-André; **Fielding, Drummond B.**, [Simulations of jet heating in galaxy clusters: successes and challenges](#), *MNRAS*, 483, 2465, 2019 (arXiv:1805.06461) [62 citations]
- 3 Stern, Jonathan; Faucher-Giguère, Claude-André; Hennawi, Joseph F.; Hafen, Zachary *et al.* (incl. **DBF**), [Does Circumgalactic O VI Trace Low-pressure Gas Beyond the Accretion Shock? Clues from H I and Low-ion Absorption, Line Kinematics, and Dust Extinction](#), *ApJ*, 865, 91, 2018 (arXiv:1803.05446) [57 citations]
- \*2 Offner, Stella S. R.; Dunham, Michael M.; Lee, Katherine I.; Arce, Héctor G. *et al.* (incl. **DBF**), [The Turbulent Origin of Outflow and Spin Misalignment in Multiple Star Systems](#), *ApJ*, 827, L11, 2016 (arXiv:1606.08445) [117 citations]
- 1 Schlieder, Joshua E.; Lépine, Sébastien; Rice, Emily; Simon, Michal *et al.* (incl. **DBF**), [The Na 8200 Å Doublet as an Age Indicator in Low-mass Stars](#), *AJ*, 143, 114, 2012 (arXiv:1202.4191) [68 citations]